G8-01-CX-stretching-algebra-kill-test-v1: Algebraic structure of W

ZFL-CX

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1 Frozen input

$$\boxed{G7 = \text{FROZEN}}$$

$$\boxed{\mathcal{C}_{\text{PROVEN}} = \{C_E^{weak}, C_{L^3}, C_F^{conc}, C_I^{freq}, C_J^{neg}\}}$$

$$\boxed{T^* < \infty \Rightarrow \int_0^{T^*} W_+ = +\infty,\ W_+ = \left(\int \omega^T S \omega\right)_+}$$

2 Mission G8-01

$$\boxed{\mathcal{C}_{\text{PROVEN}} \overset{?}{\Rightarrow} \int_0^{T^*} W_+ < \infty \text{ or } \perp}$$

Target A: $W_+ \leq F(X_k)$, $\int_0^{T^*} F < \infty \Rightarrow$ OPEN$\to$PROVEN
Target B: $\mathcal{C}_{\text{PROVEN}} \Rightarrow \perp$
Target C: KILLED if only dimensional estimate

3 Object

$W = \int_{\mathbb{R}^3} \omega_i S_{ij} \omega_j \, dx$, $S = \frac{1}{2}(\nabla u + \nabla u^T)$, $\omega = \nabla \times u$. Decomposition: $\omega = |\omega|\xi$, $W = \int |\omega|^2 \alpha$, $\alpha = \xi^T S \xi$.

4 Barriers imposed at t=0

- No $\|S\|_{L^\infty} < \infty$ without proof.
- No $|\omega^T S \omega| \leq |\omega|^2 \|S\|$ as bridge to integrability.
- No $C_{L^3} : \|u\|_{L^3} \to \infty \Rightarrow$ claim on W_+.
- No dimensional estimate counts: $\boxed{\text{no } |W| \lesssim \|u\|_3 \|\nabla\omega\|_2^2 \text{ as bridge}}$.

5 Attempt at algebraic bridge

Try orthogonal decomposition $S = S^+ - S^-$, $S^\pm \geq 0$. Then $W = W^+ - W^-$, $W^+ = \int \omega^T S^+ \omega$. $W_+ \leq W^+$ but $W^+ \leq \int |\omega|^2 |S|$.

Paraproduct: $\omega_i S_{ij} \omega_j \sim \sum_N P_N \omega \cdot S_{\sim N} \cdot P_{\sim N} \omega$.

Attempt $W_+ \lesssim \|S\|_{BMO}\|\omega\|_2^2$ or $\|S\|_{L^3}\|\omega\|_{L^3}^2$ or $\|u\|_{BMO^{-1}}\|\nabla\omega\|_2^2$. All reduce to known: $\|S\|_{L^3} \sim \|\nabla u\|_{L^3}$, $\|\omega\|_{L^3} \sim \|\nabla u\|_{L^3}$.

Hence $\boxed{W_+ \lesssim \|u\|_{L^3}\|\nabla\omega\|_2^2 \text{ up to constants}}$ inevitable without new structure.

6 Test against $\mathcal{C}_{\text{PROVEN}}$

C_{L^3} allows $\|u(t)\|_{L^3} \to \infty$. $C_F^{conc} + C_I^{freq}$ allow $N_c(t) \to \infty$ so $\|\nabla\omega\|_2$ not controlled. Thus $F(t) = \|u\|_3\|\nabla\omega\|_2^2$ satisfies $\int_0^{T^*} F = +\infty$ compatible with $\mathcal{C}_{\text{PROVEN}}$.

No algebraic identity $S = \nabla_{\text{sym}}u$ gives cancellation for $(\omega^T S\omega)_+$ beyond trace-free property $\operatorname{tr} S = 0$. Trace-free gives $\lambda_1 + \lambda_2 + \lambda_3 = 0$, but does not bound positive part: can have $\lambda_1 \gg 1$, $\lambda_2 + \lambda_3 = -\lambda_1$, α_+ still large if ξ aligns with λ_1 eigenvector. C_J^{neg} says alignment can be lost, but not quantified into integrable bound.

7 Outcome

$$\boxed{\text{KILLED}}$$

No $W_+ \leq F$ with $\int F < \infty$ from pure algebra.

Ends in dimensional estimate $\|u\|_3\|\nabla\omega\|_2^2$ which is allowed to diverge under $\mathcal{C}_{\text{PROVEN}}$.

$\mathcal{C}_{\text{PROVEN}} \not\Rightarrow \perp$ remains.

KILL conditions met: (i) only $|\omega|^2\|S\|$, (ii) needs $\|S\|_\infty$, (iii) F not integrable, (iv) no new structure beyond trace-free.